Homework 6: Rank and Inverses

EECS 245, Fall 2026 at the University of Michigan

due Sunday, November 1st, 2026 at 11:59PM Ann Arbor Time

Except for the video link in Problem 1, write your solutions to the following problems either by writing them on a piece of paper or on a tablet and scanning your answers as a PDF. Note that you are not allowed to use LaTeX, Google Docs, or any other digital document creation software to type your answers. Homeworks are due to Pensive by 11:59PM on the due date. See the **syllabus** for details on the slip day policy.

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should always explain and justify your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Before proceeding, make sure you're familiar with the **collaboration policy**.

Total Points: $10 + 8 + 17 + 5 + 8 + 3 + 12 + 9 + 6 + 14 + 8 = 100$

Problem 1: Homework 5 Concept Video (10 pts)

Record a **2- to 5-minute video** 🎥 answering **one** of the prompts below. These prompts are related to the ideas in Homework 5.

1. If you swap two columns of an orthogonal matrix, is the resulting matrix still orthogonal, and why?
2. If you append to a matrix a new column equal to the sum of its first two columns, how can you update its CR decomposition, and does its rank change?
3. If P projects onto a nonzero vector and R reflects across that same vector, does $PR = P$, $PR = R$, or neither, and why?
4. Can a matrix that projects onto a nonzero vector in $\mathbb{R}^2$ be orthogonal, and why or why not?

The video should be **just you speaking into the camera**. Your face must be visible while you are talking for the entire video. Do not include slides, a screen recording, written work, or any other visuals.

Submit your video link to the separate assignment on Pensive called **Homework 6, Problem 1 Video Link**. You can upload your video to YouTube as unlisted, share a Zoom recording link, or share a Google Drive link; whatever works. Make sure the course staff can view the video using that link.

Your submission time for Homework 6 is the later of the two submission times: your Problem 1 video link and your regular Homework 6 PDF.

This problem is graded on completion. Record a video of the required length and make an honest effort to explain the question, and you will receive full credit. Your explanation does not need to be polished or perfect.

Problem 2: Rank-Nullity Practice (8 pts)

In each part, identify whether the statement is true or false, and explain why.

a) (2 pts) There exists a 4×5 matrix A with $\text{rank}(A) = 4$ and $\dim(\text{colsp}(A)) = 3$.

b) (2 pts) There exists a 4×5 matrix B with $\text{rank}(B) = 3$ and $\dim(\text{nullsp}(B)) = 2$.

c) (2 pts) There exists a 4×5 matrix C with $\dim(\text{nullsp}(C)) = 4$ and $\dim(\text{nullsp}(C^T)) = 1$.

d) (2 pts) There exists an invertible 3×3 matrix D with $\dim(\text{nullsp}(D^T)) = 1$.

Problem 3: Spaces (17 pts)

a) (4 pts) Find a matrix A such that

$$\text{nullsp}(A) = \text{span}\left(\left\{\begin{bmatrix}2\\1\\0\\1\end{bmatrix}\right\}\right)$$

What is $\text{rank}(A)$?

b) (4 pts) Find a matrix A such that

$$\text{nullsp}(A) = \text{span}\left(\left\{\begin{bmatrix}2\\2\\1\\0\end{bmatrix}, \begin{bmatrix}3\\1\\0\\1\end{bmatrix}\right\}\right)$$

What is $\text{rank}(A)$?

c) (5 pts) Find a matrix A such that

$$\begin{bmatrix}1\\1\\5\end{bmatrix} \in \text{colsp}(A), \quad \begin{bmatrix}0\\3\\1\end{bmatrix} \in \text{colsp}(A), \quad \begin{bmatrix}1\\1\\2\end{bmatrix} \in \text{nullsp}(A)$$

What is $\text{rank}(A)$?

d) (4 pts) Let $\vec{u} = \begin{bmatrix}1\\3\\4\end{bmatrix}$ and $\vec{v} = \begin{bmatrix}8\\-2\\3\end{bmatrix}$. Explain why there **does not** exist a matrix A such that

$$\vec{u} \in \text{colsp}(A), \quad \vec{v} \in \text{nullsp}(A^T)$$

and propose one change we could make to $\vec{v}$ that would allow such an A to exist.

*Hint: In the **Orthogonal Complements example in Chapter 5.4**, and in **this video**, we discuss a fact related to this question. If you want to make a claim about the required relationship between $\vec{u}$ and $\vec{v}$, you need to re-prove it here. This shouldn't take too many lines of work.*

Problem 4: Finding Inverses (5 pts)

Suppose $A = BC$, where B is an unknown invertible 2×2 matrix and

$$A = \begin{bmatrix} 1 & 5 \\ 0 & 3 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$$

Find B^{-1}. For full credit, you may only use the formula for the inverse of a 2×2 matrix **once**; you do not need to find B first.

Problem 5: Inverse of a Rotation (8 pts)

As we've seen, the matrix

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

rotates vectors in $\mathbb{R}^2$ counterclockwise by θ radians.

a) (2 pts) Find R^{-1}.

b) (2 pts) Now, find R for a rotation of θ radians clockwise, i.e. for $-\theta$. What do you notice, and why?

c) (4 pts) A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ first rotates vectors counterclockwise by $30°$, then multiplies their horizontal coordinates by 5 and their vertical coordinates by $\frac{1}{2}$. Find the matrix of T^{-1}. *Hint: While it's possible to do this by first finding the matrix for T, that is not necessary!*

Problem 6: Trickster (3 pts)

Find a matrix A that is **not equal to the identity matrix**, but where $A^6 = I$.

Once you've found your answer, explain how you found it and verify in Python that $A^6 = I$. You can do this either in the Terminal (as described in **this video**), or in the supplemental notebook for this homework, linked **here** on DataHub or **here** in the course GitHub repository. Include a screenshot of your Python code and output.

Problem 7: Numbers of Solutions (12 pts)

Recall that if A is an $n \times d$ matrix, $\vec{x} \in \mathbb{R}^d$, and $\vec{b} \in \mathbb{R}^n$, then

$$A\vec{x} = \vec{b}$$

is a system of n equations in d unknowns, where the unknowns are the components of $\vec{x}$, i.e. x_1, x_2, ..., x_d. Solving this system is equivalent to writing $\vec{b}$ as a **linear combination of the columns of** A.

In each part, do two things:

1. Construct a matrix A for which the number of solutions (that is, number of valid $\vec{x}$'s) to the system $A\vec{x} = \vec{b}$ is the number provided.
2. Determine whether the function $f(\vec{x}) = A\vec{x}$ is one-to-one, onto, both, or neither. (See Chapter 6.2 for a refresher on the definitions.)

The first part has been done for you as an example.

a) 0 or 1, depending on $\vec{b}$

1. If there are either 0 or 1 solutions, we know that A's columns must be linearly independent. This is because if a given set of vectors is linearly independent, then any linear combination of them can only be written in one way (a fact that we proved in Chapter 4.2). A's columns must also **not** span all of $\mathbb{R}^n$, since there are some $\vec{b} \in \mathbb{R}^n$ with no solutions for $\vec{x}$, so A must have fewer columns than rows.

   One possible A is $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$. For example, $\vec{b} = \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}$ only has a single solution for $\vec{x}$, which is $\vec{x} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$, while $\vec{c} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ has no solution for $\vec{x}$.
2. The function $f(\vec{x}) = A\vec{x}$ is one-to-one, but not onto. It is one-to-one because of the fact that any linear combination of A's columns can only be written in one way, so if $\vec{x} \neq \vec{y}$, then $A\vec{x}$ and $A\vec{y}$ must also be different. It is not onto, since there are vectors in $\mathbb{R}^3$ (like $\vec{c}$ above) that aren't the output of $f(\vec{x})$.

b) (4 pts) ∞, no matter what $\vec{b}$ is

c) (4 pts) 0 or ∞, depending on what $\vec{b}$ is

d) (4 pts) 1, no matter what $\vec{b}$ is

Problem 8: Diagonal and Triangular Matrices (9 pts)

a) (2 pts) A square matrix is **diagonal** if all entries off the main diagonal are zero, as in the matrix below.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \qquad \vec{b} = \begin{bmatrix} 6 \\ 6 \\ 8 \end{bmatrix}$$

Write $A\vec{x} = \vec{b}$ as a system of equations using the variable names x_1, x_2, and x_3, and solve it by hand.

b) (3 pts) A square matrix is **upper triangular** if all entries below the main diagonal are zero, as in the matrix below.

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & -3 & 2 \\ 0 & 0 & 4 \end{bmatrix}, \qquad \vec{b} = \begin{bmatrix} 13 \\ 12 \\ 12 \end{bmatrix}$$

Write $A\vec{x} = \vec{b}$ as a system of equations using the variable names x_1, x_2, and x_3, and solve it by hand.

c) (4 pts) Suppose T is an $n \times n$ upper triangular matrix whose diagonal entries are all nonzero.

(i) Explain why $T\vec{x} = \vec{b}$ has exactly one solution for every $\vec{b} \in \mathbb{R}^n$.

(ii) What is the rank of T?

(iii) A square matrix is **lower triangular** if all entries above the main diagonal are zero. Would the same conclusions hold if T were lower triangular instead?

Problem 9: Invertibility of XX^T (6 pts)

In **Chapter 5.4**, and in **this video**, we proved that $\text{rank}(X) = \text{rank}(X^TX)$.

Here, you'll explore the rank of XX^T. Note that X^TX is a matrix containing the dot products of all pairs of X's **columns**, while XX^T is a matrix containing the dot products of all pairs of X's **rows**.

Suppose X is an $n \times d$ matrix, and XX^T is invertible. Find and explain **all** inequalities that **must** be true between n, d, and $r = \text{rank}(X)$.

Problem 10: Sherman-Morrison Inverse (14 pts)

In EECS 280 or EECS 281, you may have learned about **memoization**, which involves storing results of an earlier calculation to help speed up future calculations. This problem involves something similar.

Suppose we have an $n \times n$ matrix A whose inverse, A^{-1}, we already know. Remember that finding inverses in general is a difficult task, so once we've found one, we'd like to avoid having to invert again.

And, suppose that we need to know the inverse of

$$B = A + \vec{u}\vec{v}^T$$

which is the sum of A and a rank 1 matrix created by taking the "outer product" of $\vec{u}, \vec{v} \in \mathbb{R}^n$ (as discussed in **Chapter 5.3**). Think of B as a small update to A.

The Sherman-Morrison formula states that

$$B^{-1} = (A + \vec{u}\vec{v}^T)^{-1} = A^{-1} - \frac{A^{-1}\vec{u}\vec{v}^T A^{-1}}{1 + \vec{v}^T A^{-1}\vec{u}}$$

The formula allows us to find the inverse of $A + \vec{u}\vec{v}^T$ just by knowing $\vec{u}$, $\vec{v}$, and A^{-1}, meaning we don't need to recompute the inverse!

a) (3 pts) To start, we'll consider a simpler case of the Sherman-Morrison formula, where $A = I$, the identity matrix. Then, since $I = I^{-1}$, the matrix we're inverting is

$$B = A + \vec{u}\vec{v}^T$$

and its inverse is

$$B^{-1} = (I + \vec{u}\vec{v}^T)^{-1} = I - \frac{\vec{u}\vec{v}^T}{1 + \vec{v}^T\vec{u}}$$

B is invertible, **except when** the denominator of the fraction above is 0, i.e. $1 + \vec{v}^T\vec{u} = 0$. When $1 + \vec{v}^T\vec{u} = 0$, what is true about B?

Hint: Evaluate $B\vec{u}$. What does the result tell you about nullsp(B)?

b) (4 pts) Prove that as long as $1 + \vec{v}^T\vec{u} \neq 0$, that

$$(I + \vec{u}\vec{v}^T)\left(I - \frac{\vec{u}\vec{v}^T}{1 + \vec{v}^T\vec{u}}\right) = I$$

(Yes, this involves a fair bit of algebra.)

c) (3 pts) Now, let's return to the full-fledged Sherman-Morrison formula, where

$$B = A + \vec{u}\vec{v}^T, \qquad B^{-1} = (A + \vec{u}\vec{v}^T)^{-1} = A^{-1} - \frac{A^{-1}\vec{u}\vec{v}^T A^{-1}}{1 + \vec{v}^T A^{-1}\vec{u}}$$

Open the **supplemental Jupyter Notebook** we've created for Homework 6, which can either be found **here** on DataHub, or **here** in the course GitHub repository; also watch **this video** first with tips on using `numpy` for linear algebra.

There, you're asked to implement the Sherman-Morrison formula, and run some experiments to quantify how much quicker using the formula is than computing the inverse of B from scratch.
This problem is **not autograded**. Rather, in your submission to this part, include screenshots of your implementations of functions `generate_random_data`, `invert_B_directly`, `invert_B_with_sherman_morrison`, `run_one_experiment`, and `many_experiments_mean_sd`, along with their outputs on the provided examples.

d) (2 pts) Include screenshots of the code you used to call `many_experiments_mean_sd` for the values provided in the question, the outputs of the print statements you were asked to add, and the `plotly` line chart you were asked to create.

e) (2 pts) Answer the question posed in the Problem 10(e) section of the supplemental Jupyter Notebook. (This shouldn't be a screenshot; just write your answer in this PDF the same way you answered the other written problems.)

If you're curious, look into **Low-Rank Adaptions (LoRA)**, a relatively recent development in large language model research! The general idea is the same as we've worked with here.

Problem 11: Canva245 (8 pts)

Your goal here is to build a very basic image editing app, called **Canva245**, using what you've learned about linear transformations and matrix multiplication.

Open the **supplemental Jupyter Notebook** we've created for Homework 6, which can either be found **here** on DataHub, or **here** in the course GitHub repository; also watch **this video** first with tips on using `numpy` for linear algebra.

You won't need to submit the notebook anywhere. To get credit, complete all four tasks in the notebook and include the requested screenshots in your PDF submission to Homework 6 on Pensive, specifically under **Problem 11**. The notebook specifies what to submit for each task:

1. **Task 1 (2 pts):** Sepia.
2. **Task 2 (2 pts):** Color tint.
3. **Task 3 (2 pts):** Shearing.
4. **Task 4 (2 pts):** Rotation.